

Physical change

- doesn't produce any new chemical substances
- often easy to reverse

e.g making a mixture or dissolving a solute

Chemical change

- produces new chemical substances
- product has different properties to their reactant.

- very difficult to reverse

- mostly exothermic

e.g color change / bubbles of gas / precipitate

Rates of reaction factors ::

→ concentration

more concentration = faster rate of reaction

→ pressure

more pressure = faster rate of reaction

→ temperature

higher temp. = faster rate of reaction

→ surface area (of solid reactants)

more surface area = faster rate of reaction

→ catalyst (including enzyme)

presence of catalyst = faster rate of reaction

* a catalyst speeds up the rate of a reaction without being consumed by it

* a catalyst decreases the activation energy, E_a , of a reaction

Collision theory

- for a reaction to occur, particles must collide
- not only that, the collision must have sufficient energy to break bonds
- minimum energy that colliding particles need is known as activation energy (E_a)

Investigating the rate of reaction:

- we need to measure either how quickly the reactants are used up or how quickly the products are formed.
- practical methods
 - measure change in mass
 - measure volume of gas produced
 - measure with / without catalyst

Graphs

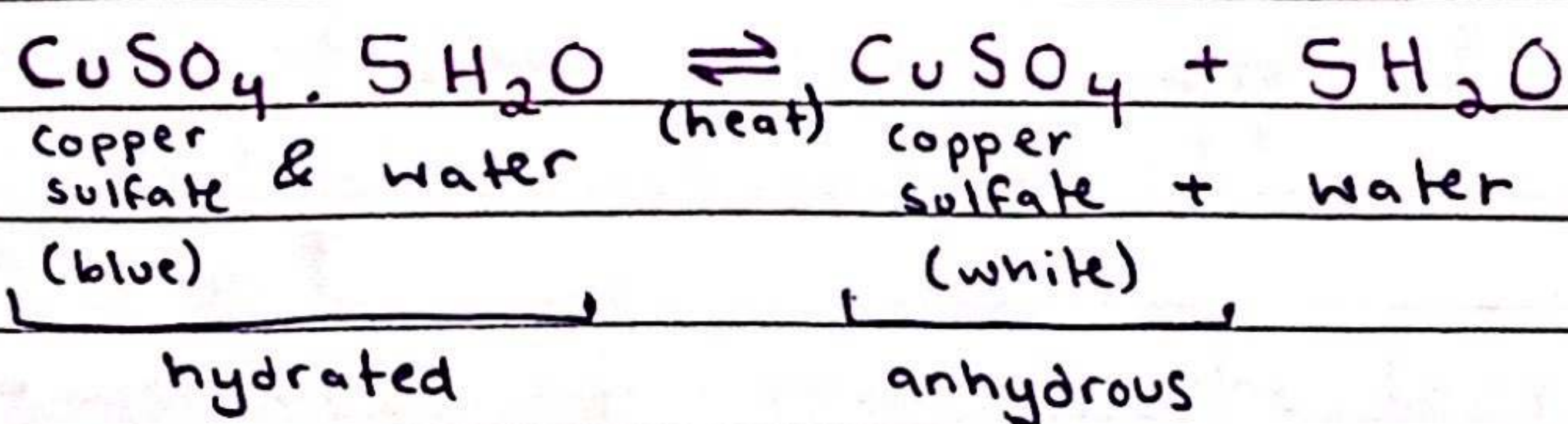
- steeper the curve, faster the rate of reaction
- rate is quickest at the beginning of the reaction
- as reaction progresses, concentration of reactants decreases thus rate decreases
- when one of the reactants is used up, the reaction stops.

Reversible reactions

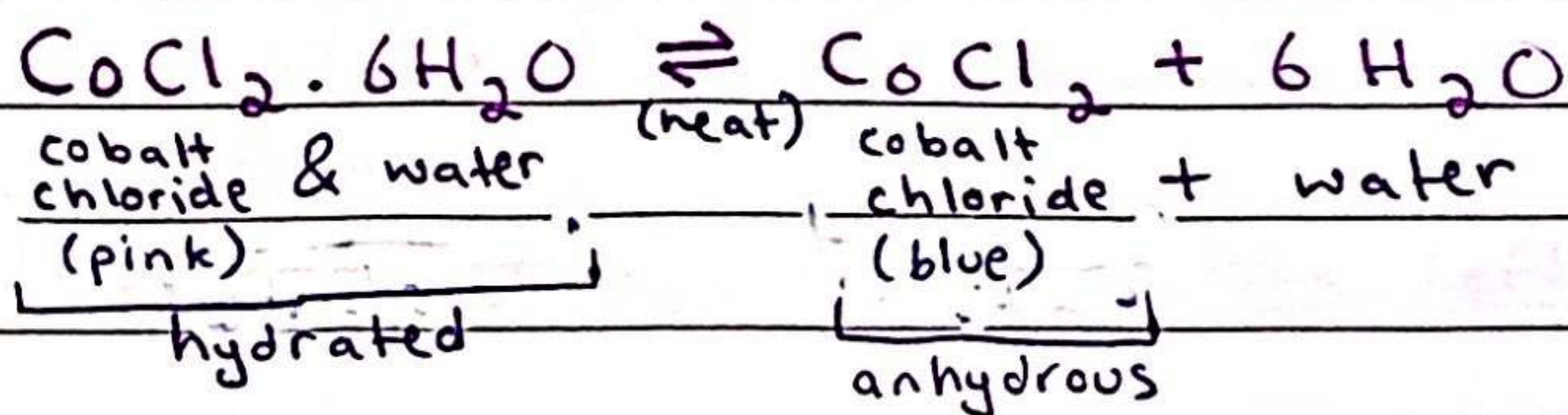
- some chemical reactions are reversible
- the symbol used is " $\rightleftharpoons$ "

conditions ::

- effect of heat on hydrated compounds
- addition of water to anhydrous compounds



exothermic $\longleftrightarrow$ endothermic



a reversible reaction in a closed system is at equilibrium when:

- * rate of forward reaction = rate of reverse reaction
- * concentrations of reactants & products is no longer changing.

Equilibrium

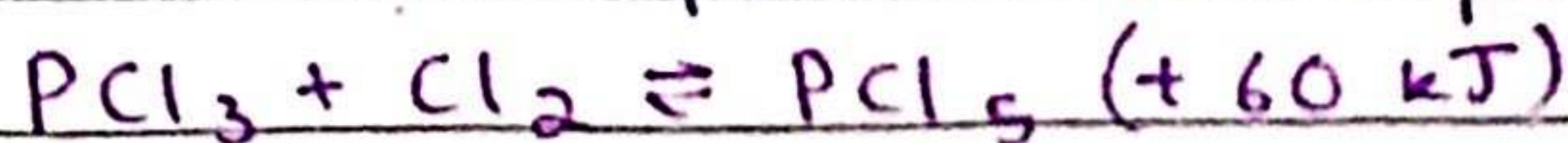
- dynamic; molecules on the left & right of the equation are changing into each other at the same rate
- concentration of the reactants & products remain same given there is no change in temp. or pressure
- only occurs in a closed system

in a reversible reaction, how is the position of equilibrium affected by:

- (1) change in temperature
- (2) change in pressure
- (3) change in concentration
- (4) use of catalyst

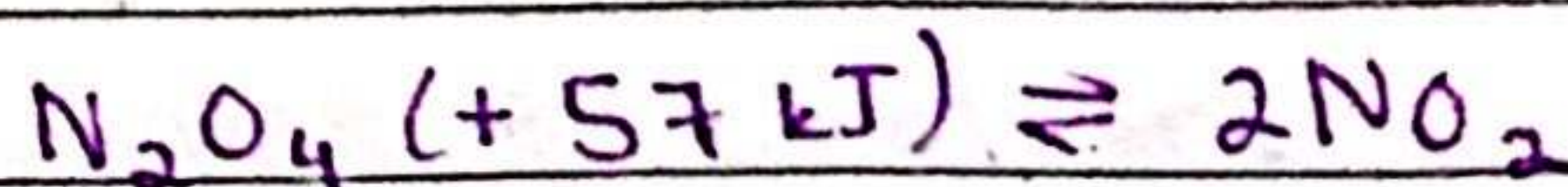
- change in temperature

* think of temperature as a part of the reaction



→ temp. ↑

↳ position of equilibrium to the left

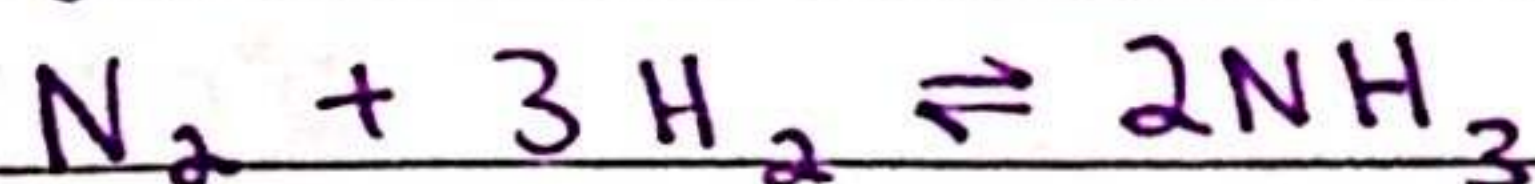


→ temp. ↓

↳ position of equilibrium to the left

- change in pressure

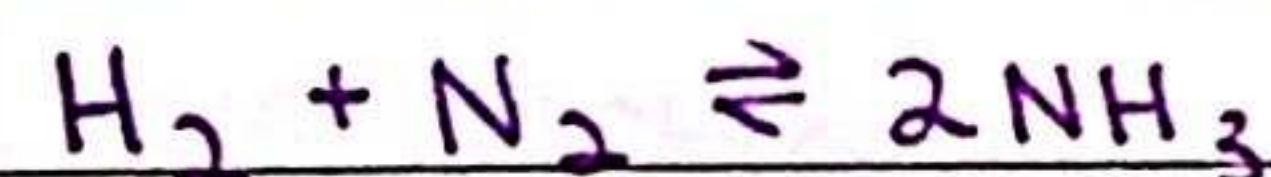
* when pressure is increased, the equilibrium shifts to the side with the fewer number of moles of gas



→ pressure ↑

↳ position of equilibrium to the right

- change in concentration



→ concentration ↑ (N_2)

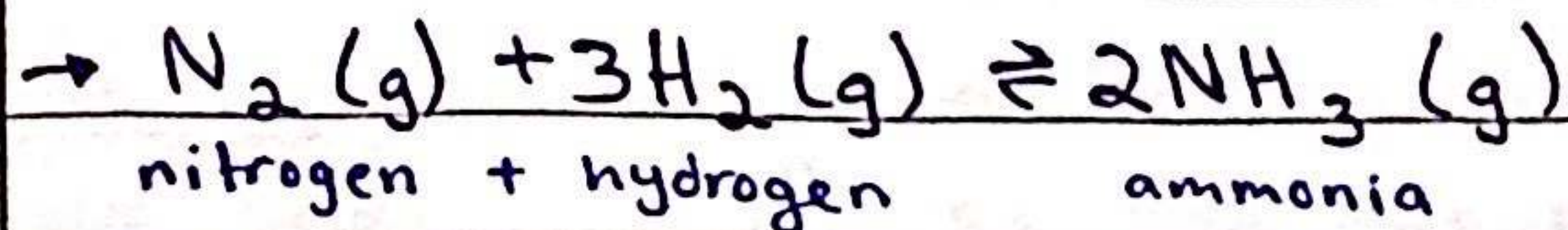
↳ position of equilibrium to the right

- use of catalyst

* doesn't affect position of equilibrium, helps reach equilibrium faster

Haber process

- used for the manufacture of Ammonia



Conditions:

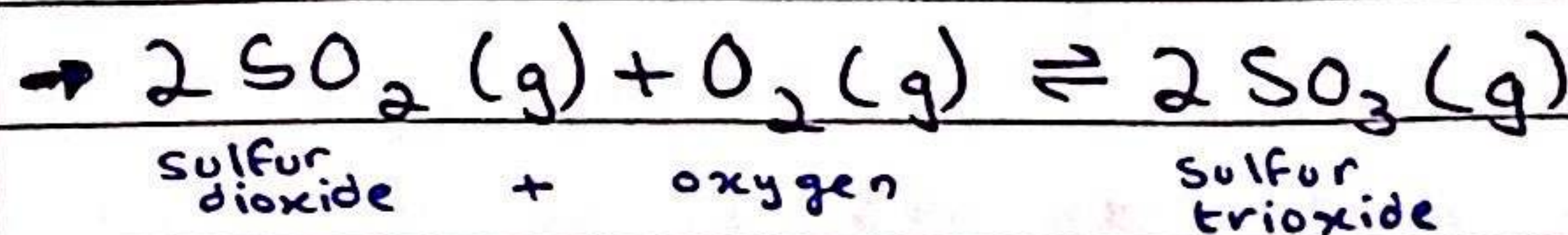
- 450°C
- 20 000 kPa / 200 atm
- iron catalyst

* the nitrogen is obtained by separating it from the air

* the hydrogen is obtained from natural gases such as methane

Contact process

- used to make sulfuric acid



turning sulfur dioxide to trioxide for the contact process

conditions:

- 450°C
- 200 kPa / 2 atm
- vanadium (V) oxide catalyst

Explaining these conditions

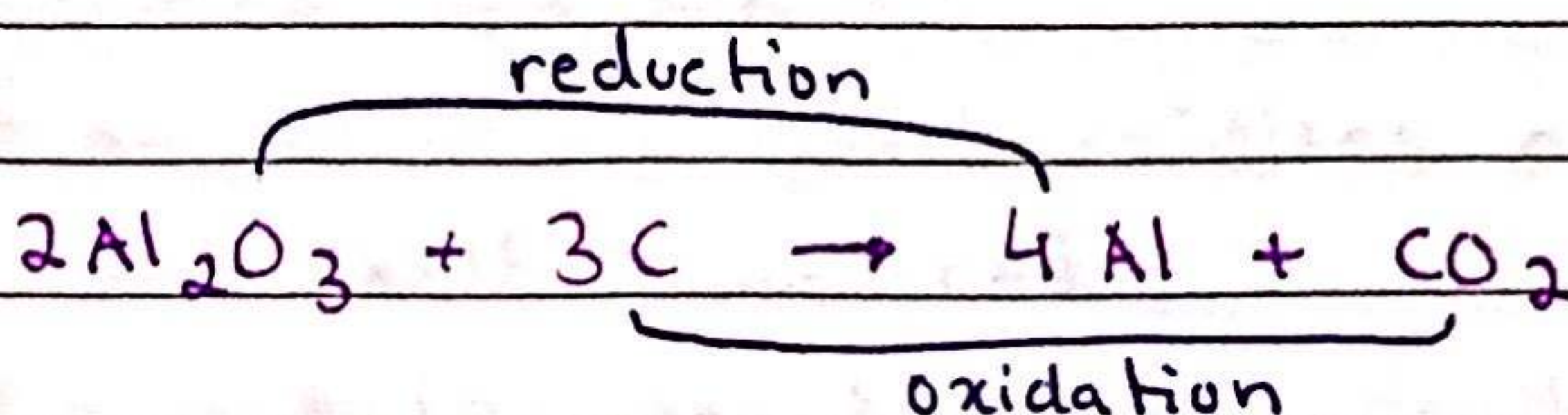
- Haber process

* 450°C temperature because it moves the equilibrium to the right, making more ammonia but since it is a low temperature, the rate of reaction is slower. So a compromise is made.

- * 20 000 kPa / 200 atm is used but it is dangerous and it is very expensive but more product is made since it's indeed a lot of pressure.
- * iron catalyst is used because it's cheap, durable and lowers " E_a " so the process can occur at lower temperatures.
- Contact process
- * 450°C temperature is used because the reaction is exothermic so that a low temperature would produce more sulfur trioxide (SO_3)
- * 200 kPa / 2 atm is used because the equilibrium position is already far to the right so there is no need for high pressure, not too expensive either
- * vanadium (V) oxide catalyst is used since it increases the rate of reaction thus reaching the equilibrium position faster.

Redox reaction

- a reaction in which oxidation and reduction happen at the same time.
- oxidation is a reaction in which oxygen is added.
- reduction is a reaction in which oxygen is removed.
- involve loss/gain of oxygen and electrons



Roman numerals for oxidation number:

$ZnO \rightarrow$ zinc oxide

$CrO \rightarrow$ chromium (II) oxide

$Cr_2O_3 \rightarrow$ chromium (III) oxide

$CrO_3 \rightarrow$ chromium (VI) oxide

* oxidation is the loss of electrons and increase of oxidation number

* reduction is the gain of electrons and decrease of oxidation number

Changes in oxidation number:

→ oxidation number of elements in their uncombined state is 0

→ oxidation number of a monatomic ion is the same as the charge on the ion

→ sum of oxidation numbers in a compound is 0

→ sum of oxidation numbers in an ion is equal to the charge on the ion.

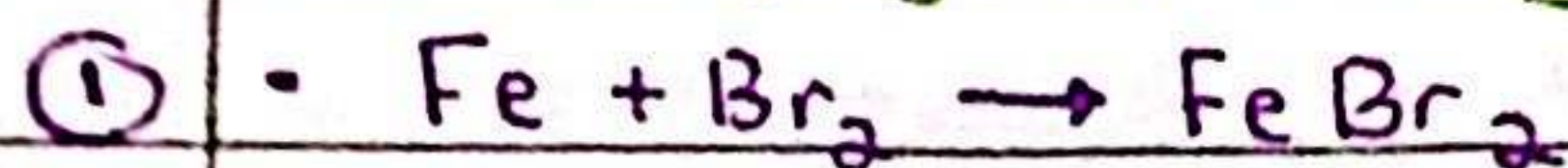
Color changes:

• so potassium manganate (VII) is an oxidising agent that when added to reducing agent, its purple color changes to colorless.

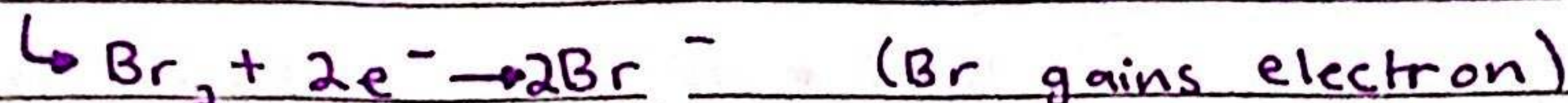
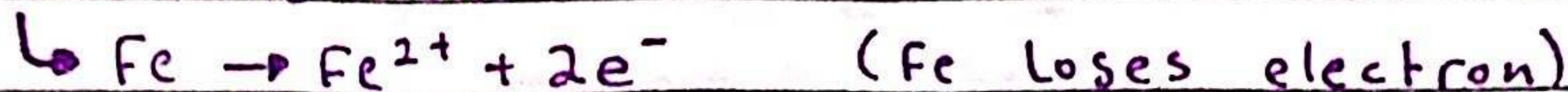
• potassium iodide (aq) is a reducing agent that when added to an oxidising agent, turns the solution red-brown.

→ an oxidising agent oxidises another substance and itself is reduced

→ a reducing agent reduces another substance and itself is oxidised

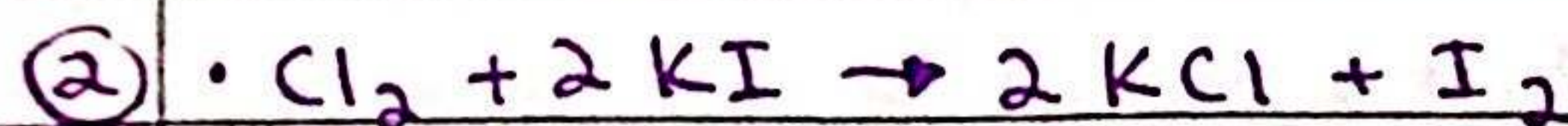
Identifying oxidising/reducing agent:

• half equations

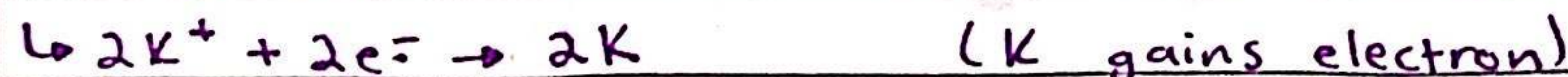


• Fe = oxidised

• Br = reduced

• reducing agent = Fe since it oxidised itself and caused Br₂ to get reduced

• half equations



• Cl = reduced

• K = N/A

• I = oxidised

• oxidising agent = Cl since it reduced itself and caused I to get

O

I

L

R

I

G

oxidation is loss reduction is gain

of electrons